valve / NRV) that opens only in the upward direction.

Strainer

The strainer is used to filter the unwanted particle present in the water to prevent the centrifugal pump from blockage.

Delivery Pipe

It is a pipe whose one end is connected to the outlet of the pump and other end goes up to the required height where water is to be delivered.

Prime mover

It is an electric motor or oil engine to drive the pump.

Working

As the electric motor starts rotating, it also rotates the impeller. The rotation of the impeller creates suction at the suction pipe. Due to suction created, the water from the sump starts coming to the casing through the eye of the impeller.

From the eye of the impeller, due to the centrifugal force acting on the water, the water starts moving towards the outer of casing.

Because of the impeller rotating at high velocity, it also rotates the water around it in the casing. As the area of the casing increases gradually in the direction of rotation, the velocity of the water decreases and the pressure increases at the outlet of the pump. Here the pressure is maximum.

Now from the outlet of the pump, the water goes to its desired location through the delivery pipe.

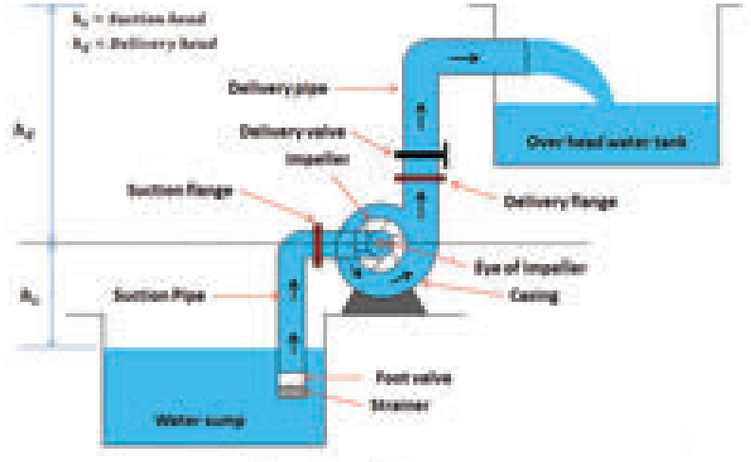
Uses

The centrifugal pump is used to raise the liquid from lower level to higher level. They are mostly used at home for filling overhead water tanks, in industries, for irrigation, etc.

7.6.7 Priming of pump

It is the process in which the suction pipe, casing and delivery pipe upto the delivery valve is filled completely with liquid to be raised from outside source before starting the motor. Priming is done to remove the air from the pump.

If air is not removed from the pump a small negative pressure created at the suction pipe doesnot allow to suck the water. So it is advised to fill the pump with water before starting it.



Centrifugal Pump

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